

HEMASHREE B M

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CAREER OBJECTIVE: EMBEDDED SYSTEMS & FIRMWARE INTERN

Detail-oriented Electronics & Communication Engineering undergraduate at Alliance University focused on Embedded Systems, Firmware Development, and Microcontroller interfacing. Hands-on expertise in bare-metal Embedded C, GPIO, timers, interrupts, UART, SPI, and hardware bring-up. Seeking an embedded engineering internship to build robust, deterministic firmware solutions.

EDUCATION

• Bachelor of Technology (B.Tech) — Electronics & Communication Engineering (ECE)

Alliance College of Engineering and Design, Alliance University, Bengaluru | **2024–2028** | *Current CGPA: 7.50 / 10.0*

Relevant Coursework: Electronic Circuits, Digital Logic & VHDL, Signals & Systems, DSP, Microprocessors, Computer Networks, Transmission Lines.

• Class XII (Science) — SDC PU College, Mulbagal | 85.55% | Class X (SSLC) — St. Anne's High School, Mulbagal | 77.76%

TECHNICAL PROFICIENCIES

Domain Competencies: Embedded C, C, Arduino Uno/Nano, ESP32, GPIO, Timers & Interrupts, UART / SPI / I2C, Sensor Interfacing, Hardware Debugging

Languages & Software: Embedded C, C, Python, Dart, VHDL, MATLAB, SQL, Flutter, Supabase, Git, Linux

Microcontrollers & Tools: ESP32, Arduino Uno/Nano, 8086, HC-05 Bluetooth, Sensors & Relays, Tinkercad, PlatformIO

VERIFIED TECHNICAL & ACADEMIC PROJECTS

• Lifemate — AI Voice-First Everyday Life Companion — *Role: Lead Architect & Solo Developer*

Technologies: Flutter, Dart, Supabase, PostgreSQL, Gemini AI, Google ML Kit, Android STT/TTS, SQLite, PKCE Auth

■ Architected secure Gemini AI companion via Supabase Edge Function proxy with rate-limiting (10 req/min/UID) and prompt caps.

■ Engineered contextual Indian Government welfare scheme matching engine across 12+ verified schemes based on age, income, and district.

• Smart Blind Stick (Assistive Ultrasonic Mobility Device) — *Role: Firmware & Hardware Developer*

Technologies: Arduino Uno, HC-SR04 Ultrasonic Sensor, Embedded C, Piezo Buzzer, Vibration Motor, Threshold Logic

■ Implemented ultrasonic time-of-flight distance calculation with multi-tier proximity danger thresholds in Embedded C.

■ Designed dual-alert sensory feedback combining progressive audio tone frequencies with haptic vibration pulses.

• Bluetooth-Controlled Home Automation System — *Role: Hardware & Firmware Developer*

Technologies: Arduino Uno, HC-05 Bluetooth Module, 4-Channel Optocoupler Relay, Embedded C, UART Serial, Android

■ Developed interrupt-driven UART serial packet parser between Android client and Arduino microcontroller.

■ Designed optoisolated relay driver stage to isolate high-voltage AC mains from low-voltage logic circuitry.

• Mobile Phone RF Activity Detection System — *Role: Circuit Designer & Developer*

Technologies: Arduino Uno, RF Sniffer Circuitry, LM358 Op-Amp, Embedded C, LED & Buzzer Indicators

■ Constructed tuned LC resonant tank circuit sensitive to 900 MHz – 2.4 GHz cellular transmission burst signals.

■ Engineered high-gain LM358 operational amplifier conditioning stage to amplify microvolt-level RF induced voltages.

KEY CERTIFICATIONS & UPSKILLING

- Embedded System Design using C Programming — Maven Silicon
- Introduction to Electronics — Georgia Institute of Technology (Coursera)
- Sensors and Sensor Circuit Design — Coursework Partner
- Crash Course on Python — Google

HACKATHONS & LEADERSHIP

- **Bharatiya Antariksh Hackathon (BAH) 2026** — ISRO / Hack2skill space-tech hackathon participation (Certificate: 2026H2S06BAH-P05872).
- **DreamFlow Buildathon 2025 (Hack2skill)** — Official Finisher; engineered and deployed functional mobile prototype.
- **6-Hour Rapid Hackathon (Alliance University)** — Technical Lead & Presenter for Bhoomi Mitra (AI agricultural intelligence pitch deck).

LANGUAGES & ATTRIBUTES

- **Languages:** Kannada (Native), English (Professional Working), Hindi, Tamil, Telugu
- **Work Attributes:** Fast Learner, Clean Code Discipline, Hardware Troubleshooting, Team Collaboration